

QUALITY ASSURANCE (QA) TEST PLAN

Project Name : Smart File System Automation and Management System

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Version : 1.0

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1. Objective

The objective of this testing activity is to verify that the Smart File System Automation and Management System functions according to the requirements specified in the Software Requirements Specification (SRS) document. Testing will ensure that automatic file classification, file naming rule validation, duplicate detection, backup mechanisms, file encryption, and report generation work accurately, securely, and reliably.

2. Scope of Testing

The following modules and functional requirements are included in testing:

1. FR-01: Automatic File Classification
2. FR-02: File Naming Rule Validation
3. FR-03: File Sorting and Viewing
4. FR-04: Automatic Backup System
5. FR-05: Duplicate File Management
6. FR-06: Recycle Bin Management
7. FR-08: File Activity History
8. FR-09: File Size Analyzer
9. FR-10: File Password Protection
10. FR-11: Favorite Files Management
11. FR-12: Duplicate File Suggestion Before Saving
12. FR-13: Smart File Search
13. FR-14: QR Code File Sharing
14. FR-15: Weekly Health Report
15. Non-Functional Requirements (Performance, Security, Reliability, Scalability)

3. Test Environment

Parameter	Details	
Operating System	Windows 10 / 11 / Linux / macOS	
Programming Language	Python 3.x	
Database	CSV files	
Core Libraries	watchdog, shutil, pathlib, hashlib, tkinter, pandas, qrcode, report	
Hardware Requirements	Intel i3 processor or higher, Min 4 GB RAM, 100 GB free storage space	

4. Functional Test Cases

Module A: Automatic File Classification

TC ID	Test Scenario	Test Steps	Expected Result	
01	Verify classification of Document files	Place .pdf, .docx, .xlsx, .pptx files into the monitored folder	Files are automatically moved to respective Documents/ sub-folders within 5 seconds.	
02	Verify classification of Media files	Place .jpg, .mp4, .mp3 files into the monitored folder	Files are moved to Pictures, Videos, and Audio folders respectively.	
03	Verify classification of Programming files	Place .c, .cpp, .py, .java, .html, .css, .js files into the folder	Files are correctly moved to respective Programming/ sub-folders.	
04	Verify handling of unsupported file extension	Place a file with an undefined extension (e.g., .xyz)	System handles it gracefully (retains in source or moves to an 'Unclassified' folder).	

Module B: File Naming Rule Validation

TC ID	Test Scenario	Test Steps	Expected Result	
01	Validate valid filename	Save/move a file matching the administrator-defined naming convention	File passes validation and is successfully stored.	
02	Detect invalid filename	Save/move a file violating the predefined naming format	Invalid filename is detected, and a suggested correct format is displayed.	
03	Prevent storage of invalid filename	Attempt to bypass naming validation with illegal characters	System blocks file storage and prompts user to rename.	

Module C: File Sorting and Viewing

TC ID	Test Scenario	Test Steps	Expected Result	
01	Sort files by alphabetical order	Apply sorting by Name (A-Z) and Name (Z-A)	Files list updates immediately in correct alphabetical order.	
02	Sort files by Type and Size	Select sorting options: File Type and File Size	Files are grouped by extension or ordered from largest to smallest/vice-versa.	

| 03 | Sort files by Date metadata | Select sorting: Date Created (Oldest/Newest First) and Date Modified | Files are accurately sorted based on their creation/modification timestamps. |

Module D: Automatic Backup System

| TC ID | Test Scenario | Test Steps | Expected Result |

| 01 | Verify folder selection and backup execution | Select a specific directory and trigger a manual/scheduled backup | Backup copy of selected folders is created successfully in the backup destination. |

| 02 | Verify incremental backup mechanism | Modify a single file inside the folder and run the backup again | Only the modified/new file is copied (Incremental backup works), saving storage. |

| 03 | Verify backup log maintenance | Check the backup logs after a backup operation concludes | Detailed logs showing backup status, source, and destination timestamps are recorded. |

Module E: Duplicate File Management

| TC ID | Test Scenario | Test Steps | Expected Result |

| 01 | Detect duplicate by Name, Size, and Hash | Put two identical files (same content/hash) with different names in the system | System accurately flags them as duplicate using File Hash comparison. |

| 02 | Remove duplicate after confirmation | Select a duplicate file, view notification, and choose 'Remove' | Duplicate file is removed safely after user confirmation, log is maintained. |

| 03 | Warning before saving a duplicate file | Attempt to save a new file that already exists in the system directory | A duplicate warning is displayed, suggesting options to overwrite or rename. |

Module F: Recycle Bin Management

| TC ID | Test Scenario | Test Steps | Expected Result |

| 01 | Verify moving deleted files to application bin | Delete a file from a monitored folder | File is moved to the application's internal recycle bin, not permanently lost. |

| 02 | Restore deleted file | Go to the Recycle Bin and select 'Restore' on a deleted item | File is restored to its original path and restore history is recorded. |

| 03 | Permanent deletion after configured duration | Set auto-delete duration to 1 minute/day and let the time lapse | Files are permanently purged automatically from the application bin. |

Module G: File Activity History

| TC ID | Test Scenario | Test Steps | Expected Result |

| 01 | Track File Creation and Modification | Create a new file and then modify its content | Separate activity logs are registered for 'File Created' and 'File Modified'. |

| 02 | Track File Movements and Renaming | Rename a file and let the system move it via FR-01 rules | Logs successfully capture 'File Renamed', 'File Moved' with Username & Timestamp. |

Module H: File Size Analyzer

| TC ID | Test Scenario | Test Steps | Expected Result |

| 01 | Analyze storage utilization metrics | Open the storage analyzer dashboard | Accurately displays largest files, largest folders, and file distribution charts. |

| 02 | Verify file count by category | Compare classified folders count with the Analyzer UI data | Category-wise file counts match precisely with actual file statistics. |

Module I: File Password Protection

| TC ID | Test Scenario | Test Steps | Expected Result |

| 01 | Encrypt file with password | Select a file, set a strong password, and apply protection | File is securely encrypted using pyzipper/cryptography backend. |

| 02 | Verify password window before access | Attempt to open the protected file | Prompt appears requesting password verification; blocks access if incorrect. |

Module J: Favorite Files & Smart Search

| TC ID | Test Scenario | Test Steps | Expected Result |

| 01 | Add/Remove Quick Access Favorites | Mark a text file as 'Favorite' and navigate to the Favorites view | File is listed under Favorites for quick access; removed successfully when unmarked |

| 02 | Advanced Search with multi-filters | Search using combinations like: Name + Extension + Date Range | Accurate search results are filtered and displayed within 2 seconds. |

Module K: QR Code Sharing & Weekly Health Report

| TC ID | Test Scenario | Test Steps | Expected Result |

| 01 | QR Code generation and link validation | Select a file and click 'Generate QR Code' | A scannable QR code is generated with a temporary local share link. |

| 02 | Automated Weekly Report Export | Trigger the automated generation of the weekly health report | Comprehensive health report is generated within 30 seconds in PDF and CSV. |

5. Non-Functional Testing (Performance & Security)

| TC ID | Test Scenario | Test Steps | Expected Result |

| 01 | Performance - Classification speed | Add a batch of 50 files simultaneously into the folder | Automated classification of all files completes within 5 seconds. |

| 02 | Security - Credential protection | Inspect database records for user authentication profiles | User passwords must be encrypted and stored securely using hashing. |

| 03 | Scalability - High file volume | Populate the system directories with over 100,000 files | System remains stable, search and watchdogs perform without memory leaks. |

6.Acceptance Criteria:

1. Files are automatically categorized into their correct folders within 5 seconds.
2. Name rule configurations block unformatted entries and display alternate naming suggestions.
3. Duplicate files are detected reliably by name, size, and cryptographic hash.
4. Folder backup captures incremental files and maintains a 99% execution success rate.
5. Password protection completely encrypts the files and asks for mandatory verification.
6. Smart search, QR code generation, and storage analysis show 100% accurate statistics.
7. The weekly summary health report is successfully generated in both PDF and CSV formats.
8. All planned QA test cases pass successfully with zero critical defects remaining